

Insurance Company ? Mathematical Reserves

Source: <https://www.prarulebook.co.uk/prarules/insurance-company-mathematical-reserves/03-03-2026>

Effective Date: 03-03-2026

Type: PRA Rule

Scraped: 2026-03-02 20:52

1

Application and Definitions

1.1

Unless otherwise stated, this Part applies to a _non-directive insurer_ which carries on _long-term insurance business_ , other than a _non-directive friendly society_.

1.2

Except for 16.3(1)) this Part does not apply to _final bonuses_.

1.3

In this Part, the following definitions shall apply:

benefits payable

means benefits payable under a _long-term insurance contract_ , including:

1. (1) all guaranteed benefits including guaranteed _surrender values_ and paid-up values;
2. (2) vested, declared and allotted bonuses to which the _policyholder_ is entitled;
3. (3) all options available to the _policyholder_ under the terms of the contract; and
4. (4) discretionary benefits payable in accordance with any relevant provisions of the _FCA Handbook_.

dividend yield

has the meaning given in 9.6")(1).

earnings yield

has the meaning given in 9.6")(2).

future premium

means the value attributed to a _premium_ due in any future _financial year_.

internal rate of return

has the meaning given in 9.7.

non-attributable expenses

means expenses which are not directly attributable to a particular _contract of long-term insurance_.

recoverable acquisition expenses

means the amount of expenses, after allowing for the effects of taxation, which it is reasonable to expect will be recovered from future premiums payable under the contract.

risk-adjusted yield

means the risk-adjusted yield calculated in accordance with 9 and 10.

running yield

has the meaning given in 9.5.

2

Basic Valuation Method

2.1

1. (1) Subject to (2), a firm must establish its mathematical reserves using a prospective actuarial valuation on prudent assumptions of all future cash flows expected to arise under, or in respect of, each of its contracts of long-term insurance.

2. (2) A firm may use a retrospective actuarial valuation where:

1. (a) a prospective method cannot be applied to a particular type of contract; or
2. (b) the firm can demonstrate that the resulting amount of the mathematical reserves would be no lower than would be required by a prudent prospective actuarial valuation.

3

Methods and Assumptions

3.1

In the actuarial valuation under 2.1, a firm must use methods and prudent assumptions which:

1. (1) are appropriate to the business of the firm ;
2. (2) are consistent from year to year without arbitrary changes;
3. (3) are consistent with the method of valuing assets;
4. (4) include appropriate margins for adverse deviation of relevant factors which are sufficiently prudent to ensure that there is no significant foreseeable risk that liabilities to policyholders in respect of contracts of long-term insurance will not be met as they fall due;
5. (5) recognise the distribution of profits (that is, emerging surplus) in an appropriate way over the duration of each contract of insurance ;
6. (6) take into account its regulatory duty to treat its customers fairly under any relevant provision of the FCA Handbook ; and
7. (7) are in accordance with generally accepted actuarial practice.

4

Valuation of Individual Contracts

4.1

1. (1) Subject to (2) and (3), a _firm_ must determine the amount of the _mathematical reserves_ separately for each _contract of long-term insurance_.
2. (2) Approximations or generalisations may be made:
 1. (a) in the case of _non-attributable expenses_ , in relation to a group of contracts with the same or similar expense risk characteristics, provided that the _mathematical reserves_ in respect of such expenses established by the _firm_ in relation to that group of contracts have a minimum value of at least zero; and
 2. (b) in any other case, where they are likely to provide the same, or a higher, result than a determination made in accordance with (1).
3. (3) A _firm_ must set up additional _mathematical reserves_ on an aggregated basis for general risks that are not specific to individual contracts.

5

Negative Mathematical Reserves

5.1

A _firm_ may calculate a negative value for the _mathematical reserves_ in respect of a _contract of long-term insurance_ provided that:

1. (1) this is based on assumptions which meet the general requirements for prudent assumptions as set out in 3.1;
2. (2) the contract does not have a _surrender value_ which at the _actuarial valuation date_ is guaranteed; and
3. (3) the total _mathematical reserves_ established by the _firm_ have a minimum value of at least:
 1. (a) where the _firm's_ contracts of long-term insurance_ include _linked long-term __contract of insurance_ , the sum of the surrender values of all its _linked long-term __contract of insurance _at the _actuarial valuation date_ ; and
 2. (b) in any other case, zero.

6

Avoidance of Future Valuation Strain

6.1

1. (1) A _firm_ must establish _mathematical reserves_ for a _contract of insurance_ which are sufficient to ensure that, at any subsequent date, the _mathematical reserves_ then required are covered solely by:
 1. (a) the assets covering the current _mathematical reserves_ ; and
 2. (b) the resources arising from those assets and from the contract itself.
2. (2) For the purposes of (1), the _firm_ must assume that:
 1. (a) the assumptions adopted for the current valuation of liabilities remain unaltered and are met; and
 2. (b) discretionary benefits and charges will be set so as to fulfil its regulatory duty to treat its customers fairly under any relevant provision of the _FCA Handbook_.
3. (3) Subject to (4), the requirements in (1) may be applied to a group of similar contracts instead of to the individual contracts within that group.
4. (4) The requirements in (1) must be applied to a group of contracts in relation to which _mathematical reserves_ in respect of _non-attributable expenses_ are established for that group of contracts in accordance with 4.1(a)(2)(a), instead of to the individual contracts within that group.

7

Cash Flows to Be Valued

7.1

In a prospective valuation, a _firm_ must:

1. (1) include in the cash flows to be valued, the following:
 1. (a) _future premiums_ ;
 2. (b) expenses, including commissions;
 3. (c) _benefits payable_ ; and
 4. (d) subject to (2), amounts to be received or paid in respect of _contracts of long-term insurance_ under contracts of _reinsurance_ or _analogous non-reinsurance financing agreements_ ; but
2. (2) exclude from those cash flows amounts recoverable from an _ISPV_.

8

Valuation Rates of Interest

8.1

In calculating the present value of future net cash flows, a _firm_ must determine the rates of interest to be used in accordance with 8.2, 9 and 10.

8.2

1. (1) The rates of interest required by 8.1 to be used by a _firm_ for the calculation of the present value of a _long-term insurance liability_ must not exceed 97.5% of the _risk-adjusted yield_ that is expected to be achieved on:
 1. (a) the assets allocated to cover that liability;
 2. (b) the reinvestment of sums expected to be received from those assets; and
 3. (c) the investment of future _premium_ receipts.
2. (2) The requirements in (1) do not apply to a _contract of long-term insurance_ in respect of which the _firm_ has calculated a negative value for the _mathematical reserves_ in accordance with 5.1.
3. (3) For the purposes of (1), the rates of interest assumed must allow appropriately for the rates of tax that apply to the investment return on _policyholder_ assets.
4. (4) For the purposes of (3), the rates of tax assumed must be such that the _firm's_ total implied liability for tax arising from the allocation of assets to liabilities is not less than the _firm's_ actual expected liability for tax for the period in respect of which tax is to be assessed.

9

Risk-Adjusted Yield

9.1

A _risk-adjusted yield_ on an asset must be calculated by:

1. (1) taking the asset together with any covering _derivatives_ , forward transactions and _quasi-derivatives_ ;
2. (2) assuming that the factors which affect the yield will remain unchanged after the valuation date;
3. (3) valuing the asset (together with any _offsetting transaction_) in accordance with Insurance Company ? Overall Resources and Valuation 3 to 8;
4. (4) making reasonable assumptions as to whether, and if so when, any options or other rights embedded in the asset (or in any _offsetting transaction_) will be exercised.

9.2

For the purpose of 9.1"(2), the factors that affect yield should be ascertained as at the valuation date (that is, the date to which present values of cash flows are being calculated). All changes known to have occurred by that date must be taken into account including:

1. (1) changes in the rental income from _real estate_ ;
2. (2) changes in dividends or audited profit on equities;
3. (3) known or forecast changes in dividends which have been publicly announced by the _issuer_ by the valuation date;
4. (4) known or forecast changes in earnings which have been publicly announced by the _issuer_ by the valuation date;
5. (5) alterations in capital structure; and
6. (6) the value (at the most recent date at or before the valuation date for which it is known) of any determinant of the amount of any future interest or capital payment.

9.3

The _risk-adjusted yield_ is either:

1. (1) for equities and _real estate_ , a _running yield_ ; or
2. (2) for all other assets, the _internal rate of return_.

9.4

The _risk-adjusted yield_ on a basket of assets is the arithmetic mean of the _risk-adjusted yield_ on each asset weighted by that asset's _market value_.

9.5

The _running yield_ :

1. (1) for _real estate_ , is the ratio of:
 1. (a) the rental income arising from the _real estate_ over the previous 12 _months_ ; to
 2. (b) the _market value_ of the _real estate_.
2. (2) for equities, is:
 1. (a) the _dividend yield_ , if the _dividend yield_ is more than the _earnings yield_ ;
 2. (b) otherwise, the sum of the _dividend yield_ and the _earnings yield_ , divided by two.

9.6

For the purposes of 9.5"(2):

1. (1) the _dividend yield_ is the ratio (expressed as a percentage) of dividend income over the previous 12 _months_ from the equities for which the _running yield_ is being calculated ("the relevant equities") to the _market value_ of those equities;
2. (2) the _earnings yield_ is the ratio (expressed as a percentage) of the audited profit (including exceptional items and extraordinary items) for the preceding _financial year_ of the _issuer_ of the relevant equities to the _market value_ of those equities;
3. (3) the _earnings yield_ must be calculated in accordance with whichever is most appropriate (to the _issuer_ of the relevant equities) of _UK_, _US_ or international generally accepted accounting practice.

9.7

The internal rate of return on an asset is the annual rate of interest which, if used to calculate the present value of future income (before deduction of tax) and of repayments of capital (before deduction of tax) would result in the sum of those amounts being equal to the market value of the asset.

9.8

In both the running yield and internal rate of return the yield must be reduced to exclude that part of the yield that represents compensation for credit risk arising from the asset.

9.9

Provision for credit risk for securities that are not credit-rated must be made on principles at least as prudent as those adopted for credit-rated securities.

10

Investment and Reinvestment

10.1

The risk-adjusted yield assumed for the investment or reinvestment of sterling sums (other than sums expected to be received within the next three years) must not exceed the lowest of:

1. (1) the higher of:
 1. (a) the long-term gilt yield; and
 2. (b) the greater of:
 1. (i) the forward gilts yield; and
 2. (ii) the forward rate on sterling interest rate swaps, reduced to exclude that part of the rate that represents compensation for credit risk;

where the forward yields and forward rates corresponding to the time when the sums are expected to be received are weighted so as to reflect the investment and reinvestment characteristics of the liabilities covered;

1. (2) 3% per annum, increased by two thirds of the excess, if any, of the percentage in (1) over 3% per annum; and
2. (3) 6.5% per annum.

10.2

The risk-adjusted yield assumed for the investment or reinvestment of those sterling sums expected to be received within the next three years must not exceed the risk-adjusted yield on the assets actually held adjusted linearly over the three-year period to the risk-adjusted yield determined under 10.1.

10.3

The risk-adjusted yield assumed for the investment or reinvestment of sums denominated in a currency other than sterling must be at least as prudent as in 10.1 and 10.2 taking into account the yields on government securities denominated in that currency.

10.4

For the purpose of 10.3 the yields on the government securities must be reduced to exclude that part of the yield that represents compensation for credit risk unless, in relation to the issuer of those securities, a credit rating is available from at least one of the following rating agencies in the corresponding rating categories;

1. (1) for A.M. Best Company, ?aaa? or ?aa?;
2. (2) for Fitch Ratings, ?AAA? or ?AA?;
3. (3) for Moody's Investors Service ?Aaa? or ?Aa?; or
4. (4) for Standard & Poor's Corporation ?AAA? or ?AA?.

11

Future Premiums

11.1

This Chapter applies to _with-profits policies_ , except _accumulating with-profits policies_ written on a recurring single premium basis.

11.2

A _future premium_ must not exceed the lower of the value of:

1. (1) the actual _premium_ payable under the contract; and
2. (2) the _net premium_ , which may be increased for _deferred acquisition costs_ in accordance with 12.1.

11.3

Where the terms of a _contract of insurance_ have changed since it was first entered into, a _firm_ must, in determining the _net premium_ for the purpose of 11.2"(2), treat that change as if either:

1. (1) it had been included in the original contract but came into effect from the time the change became effective; or
2. (2) the original contract were cancelled and replaced by a new contract (with an initial _premium_ paid on the new contract equal to the liability under the original contract immediately prior to the change); or
3. (3) subject to 11.4, it gave rise to two separate contracts where:
 1. (a) all _premiums_ are payable under the first contract and that contract provides only for such benefits as those _premiums_ could have purchased from the _firm_ at the date the change became effective; and
 2. (b) no _premiums_ are payable under the second contract and that contract provides for all the other benefits.

11.4

For the purposes of 11.3"(3), a _firm_ must not treat the change referred to in 11.3 as giving rise to two separate contracts unless a meaningful comparison can be made between the terms of the contract (as changed) and the terms upon which the _firm_ was newly _effecting contracts of insurance_ at the time the contract was changed.

12

Future Premiums: Adjustment for Deferred Acquisition Costs

12.1

1. (1) The amount of any increase to the _net premium_ for _deferred acquisition costs_ must not exceed the equivalent of the _recoverable acquisition expenses_ spread over the period of _premium_ payments and calculated in accordance with the rates of interest, mortality and morbidity assumed in calculating the _mathematical reserves_.
2. (2) The _recoverable acquisition expenses_ in (1) must not exceed the lower of:
 1. (a) the value of the excess of actual _premiums_ over _net premiums_ ; and
 2. (b) 3.5% of the _relevant capital sum_.

3. (3) _Recoverable acquisition expenses_ may be calculated as the average for a group of similar contracts weighted by the _relevant capital sum_ for each contract.

13

Future Premiums: Accumulating With-Profits Policies

13.1

This Chapter applies to _accumulating with-profits policies_ written on a recurring single _premium_ basis.

13.2

1. (1) A _firm_ must not attribute any value to a _future premium_ under the contract.
2. (2) Any liability arising only upon the payment of that _premium_ may be ignored except to the extent that the value of that liability upon payment would exceed the amount of that _premium._

14

Expenses

14.1

1. (1) A _firm_ must make provision for expenses, either implicitly or explicitly, in its _mathematical reserves_ of an amount which is not less than the amount expected, on prudent assumptions, to be incurred in fulfilling its _contracts of long-term insurance_.
2. (2) For the purpose of (1), expenses must be valued:
 1. (a) after taking account of the effect of taxation;
 2. (b) having regard to the _firm's_ actual expenses in the last 12 _months_ before the _actuarial valuation date_ and any increases in expenses expected to occur in the future;
 3. (c) after making prudent assumptions as to the effects of inflation on future increases in prices and earnings; and
 4. (d) at no less than the level that would be incurred if the _firm_ were to cease to transact new business 12 _months_ after the _actuarial valuation date_.
3. (3) A _firm_ must not rely upon an implicit provision arising from the method of valuing _future premiums_ except to the extent that:
 1. (a) it is reasonable to assume that expenses will be recoverable from _future premiums_ ; and
 2. (b) the expenses would only arise if the _future premiums_ were received.

14.2

The provisions for expenses (whether implicit or explicit) required by 14.1 must be sufficient to cover all the expenses of running off the _firm's_ existing _long-term insurance business_ including:

1. (1) all discontinuance costs (for example, redundancy costs and closure costs) that would arise if the _firm_ were to cease transacting new business 12 _months_ after the _actuarial valuation date_ in circumstances where (and to the extent that) the discontinuance costs exceed the projected surplus available to meet such costs;
2. (2) all costs of continuing to service the existing business taking into account the loss of economies of scale from, and any other likely consequences of, ceasing to transact new business at that time; and
3. (3) the lower of:
 1. (a) any projected valuation strain from writing new business for the 12 _months_ following the _actuarial valuation date_ to the extent the actual amount of that strain exceeds the projected surplus on prudent assumptions from existing

business in the 12 _months_ following the _actuarial valuation date_ ; and

2. (b) any projected new business expense overrun from writing new business for the 12 _months_ following the _actuarial valuation date_ to the extent the projected expenses exceed the expenses that the new business can support on a prudent basis.

15

Mortality and Morbidity

15.1

A _firm_ must set the assumptions for mortality and morbidity using prudent rates of mortality and morbidity that are appropriate to the country or territory of residence of the person whose life or health is insured.

16

Options

16.1

1. (1) When a _firm_ establishes its _mathematical reserves_ in respect of a _contract of long-term insurance_ , the _firm_ must include an amount to cover any increase in liabilities which might be the direct result of its _policyholder_ exercising an option under, or by virtue of, that _contract of insurance_.

2. (2) Where the _surrender value_ of a contract is guaranteed, the amount of the _mathematical reserves_ for that contract at any time must be at least as great as the value guaranteed at that time.

16.2

1. (1) Where a _policyholder_ may opt to be paid a cash amount or a series of cash payments (including the amount or amounts likely to be paid on a voluntary discontinuance), the _mathematical reserves_ for the _contract of insurance_ established under 2.1 must be sufficient to ensure that the payment or payments could be made solely from:

1. (a) the assets covering those _mathematical reserves_ ; and
2. (b) the resources arising from those assets and from the contract itself.

2. (2) For the purposes of (1), the _firm_ must assume that:

1. (a) the assumptions adopted for the current valuation remain unaltered and are met; and
2. (b) discretionary benefits and charges will be set so as to fulfil the _firm's_ regulatory duty to treat its customers fairly under any relevant provision of the _FCA Handbook_.

3. (3) (1) may be applied to a group of similar contracts instead of to the individual contracts within that group, except where the cash amount or series of cash payments is the amount or amounts likely to be paid on a voluntary discontinuance.

16.3

For the purposes of 16.2, a _firm_ must assume that the amount of a cash payment secured by the exercise of an option is:

1. (1) in the case of an _accumulating with-profits policy_ , the lower of:

1. (a) the amount which the _policyholder_ would reasonably expect to be paid if the option were exercised, having regard to the representations made by the _firm_ and including any expectations of a _final bonus_ ; and
2. (b) that amount, disregarding all discretionary adjustments;

2. (2) in the case of any other _policy_ , the amount which the _policyholder_ would reasonably expect to be paid if the option were exercised, having regard to the representations made by the _firm_ , without taking into account any

expectations regarding future distributions of profits or the granting of discretionary additions in respect of an _established surplus_.

17

Persistency Assumptions

17.1

A _firm_ must, when making assumptions about voluntary discontinuance rates in the calculation of the _mathematical reserves_, ensure that those assumptions meet the general requirements for prudent assumptions as set out in 3.1.

18

Reinsurance

18.1

In this Chapter, references to:

1. (1) _reinsurance_ and contracts of _reinsurance_ include _analogous non-reinsurance financing agreements_ ;
2. (2) reinsured risks, in relation to a contract of _reinsurance_ entered into by a _firm_, means that part of:
 1. (a) the risks insured by the _firm_ under _contracts of long-term insurance_ entered into by it; and
 2. (b) the other risks arising directly from the _firm's_ long-term insurance business;
3. that have been transferred to the _reinsurer_ under that contract of _reinsurance_; and
3. (3) _reinsurance_ cash outflows include any reduction in _policy_ liabilities recognised as covered under a contract of _reinsurance_ or any reduction of any debt to the _firm_ under or in respect of a contract of _reinsurance._

18.2

A _firm_ must value _reinsurance_ cash flows using methods and assumptions which are at least as prudent as the methods and assumptions used to value the underlying _contract of insurance_ which have been reinsured.

18.3

For purposes of 18.2:

1. (1) _reinsurance_ recoveries must not be recognised unless the underlying liabilities to which they relate have also been recognised;
2. (2) _reinsurance_ cash outflows need not to be valued provided that:
 1. (a) in accordance with 18.5 and 18.6, they are unambiguously linked to the emergence as surplus of margins included in the valuation of existing _contract of insurance_ or to the exercise by a _reinsurer_ of its rights under a termination clause; and
 2. (b) the conditions in 18.4 are satisfied;
3. (3) _reinsurance_ cash inflows that are contingent on factors or conditions other than the reinsured risks must not be valued.

18.4

The conditions referred to in 18.3(b)(2)(b) are that:

1. (1) the _reinsurance_ is not connected with any other transaction, which, when taken together with the _reinsurance_, could result in the requirements set out in 18.3(2) no longer being satisfied or in the risk transferred under the

reinsurance being undermined; and

2. (2) the present value of the future _reinsurance_ cash outflows that may be disregarded under 18.3"(2) must not at any time exceed the value of the aggregate net cash inflows that have already been received by the _firm_ under the contract of _reinsurance_ accumulated at an assumed rate of SONIA + 6% per annum.

18.5

For the purposes of 18.3"(2), the "link" must be such that a contingent liability to pay or repay the amount to the _reinsurer_ could not arise except when, and to the extent that, the margins in the valuation of the existing _contract of insurance_ emerge as surplus, or the _reinsurer_ exercises its rights under a termination clause in the contract of _reinsurance_ as a result of:

1. (1) fraudulent conduct by the _firm_ under or in relation to the contract of _reinsurance_; or
2. (2) a representation as to the existence, at or before the time the contract of _reinsurance_ is entered into, of a state of affairs which is within the knowledge or control of the _firm_ and which is material to the _reinsurer's_ decision to enter into the contract being discovered to be false; or
3. (3) the non-payment of _reinsurance_ __premiums _by the _firm_; or
4. (4) a transfer by the _firm_ of the whole or a specified part of its business without the agreement of the _reinsurer_, except where that agreement has been unreasonably withheld.

18.6

1. (1) Subject to (2), for the purposes of 18.3"(2) and 18.5, future surplus may only be offset against future _reinsurance_ cash outflow in respect of surplus on _non-profit policies_ and the charges or shareholder transfers arising as surplus from _with-profits policies_.
2. (2) Such charges and transfers may only be allowed for to the extent consistent with the regulatory duty of the _firm_ to treat its customers fairly under any relevant provision of the _FCA Handbook_.

19

Record Keeping

19.1

A _firm_ must make, and retain for an appropriate period, a record of:

1. (1) the methods and assumptions used in establishing its _mathematical reserves_ , including the margins for adverse deviation, and the reasons for their use; and
2. (2) the nature of, reasons for, and effect of, any change in approach, including the amount by which the change in approach increases or decreases its _mathematical reserves_.